EpiSurge — Experimental Results

Does pathogen genomic composition carry forecasting signal for regional influenza incidence beyond what case history already provides — and at which forecast horizons?

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Experiments E1, E2, E3a, E3b, E3c and E6 · six influenza seasons, ten US HHS regions

Summary of findings.

1. **E1.** Publication delay removes genomic coverage *specifically before the epidemic peak*: 97.1% of core region-weeks carry collection-dated composition before the peak against 34.8% under publication-dating, a 62.3-point gap, with 8–11 consecutive onset weeks at exactly zero in all six seasons.
2. **E2.** Three reference models establish the bar. Only three claims hold in every fold; the fold that crosses the COVID gap reverses sign and is reported separately rather than averaged away.
3. **E3a.** For *incidence* forecasting, subtype composition carries no recoverable signal at any horizon — oracle or as-of — against a positive control reaching +50% and a detectable-effect floor of 5.5–10.8%.
4. **E3b.** For *subtype dominance*, composition does forecast, and publication delay costs 43.5–49.8% of that skill, sign-consistent at every horizon with all 30 bootstrap intervals excluding zero. Roughly 85–95% of the damage is staleness rather than absence.
5. **E3c.** Neither obvious remedy works alone. Faster submission with today's window recovers 5.6%; a tighter window at today's submission speed is 29.2% *worse*. Together they recover 33–37%.
6. **E6.** No inter-region graph relation beats a density-matched random graph, in a GNN that demonstrably learns. The mechanism is measured: correlation structure does not persist between seasons.

1. Dataset and common methodology

All six experiments share one data foundation and one evaluation protocol, so results are directly comparable across them.

Seasons	6 — 2016-17 to 2023-24, excluding 2020-21 and 2021-22 (COVID disruption) and 2015-16 (anomalous deposit profile)
Regions	10 US HHS regions
Target	Weekly weighted ILI % (CDC FluView), final revised values
Inputs	Values as published at the forecast origin (vintage-correct)
Horizons	1, 2, 3, 4, 6, 8 weeks
Genomic records	32,141 human influenza A HA sequences, NCBI GenBank
Joined table	3,120 region-weeks × 64 columns; 1,980 in the core evaluation window
Evaluation	Rolling origin, 5 folds, expanding window
Sources joined	NCBI GenBank, CDC FluView via Delphi Epidata, ERA5 reanalysis, US Census, GADM v4.1

Table 1.1 — Dataset and scope.

1.1 Three methodological rules applied throughout

Vintage-correct inputs. At forecast origin T, only data published by T is used as input. The target uses final revised values, following the CDC FluSight scoring convention. The distinction between *collection-dated* ("oracle") and *publication-dated* ("as-of") genomic data is the axis every genomic result turns on.

Paired evaluation. Within each (fold, horizon, stratum) every model is scored on an identical row set, verified in code. Differences therefore cannot come from different denominators.

Sign consistency, not pooled means. With five folds, a t-test across folds is not meaningful. The reportable statistic is whether every fold agrees in sign; a horizon whose sign flips supports no claim and is marked **FLIP** throughout. Fold 4 (test 2022-23) crosses the COVID discontinuity and is reported separately from the four pre-COVID folds in every table.

fold	training seasons	test season
fold 1	2016-17	2017-18
fold 2	2016-17 … 2017-18	2018-19
fold 3	2016-17 … 2018-19	2019-20
fold 4	2016-17 … 2019-20	2022-23 — crosses the COVID gap
fold 5	2016-17 … 2022-23	2023-24

Table 1.2 — Rolling-origin folds. Fold 4 trains only on pre-COVID seasons and tests on the first post-COVID season.

2. E1 — Delay and coverage timing

Objective

A genomic sequence becomes usable to a forecaster only when it is *published*, not when it is *collected*. Published work that uses pathogen genomics for incidence forecasting routinely indexes sequences by collection date, which credits a forecast with data that did not exist at the moment it would have been issued. E1 measures the size and the *timing* of that difference on real submission dates.

Methodology

For each region-week, genomic coverage was computed three ways: by collection date (oracle), and by publication date under two collection windows — sequences collected 8–16 weeks before the origin, and 4–12 weeks before. Coverage was then stratified two ways, which must not be conflated: by *calendar phase* (MMWR weeks 40–52 against 1–20), and by *position relative to each season's own peak*.

Results

availability	pre-peak	peak or after
Oracle (collection-dated)	97.1	86.6
As-of, 8-16 wk window	34.8	89.6
As-of, 4-12 wk window	37.4	85.0

Table 2.1 — Core-season genomic coverage (%), row-weighted, split at each season's own peak. The oracle loses 10.5 points from before the peak to after it; publication-dating loses 62.3 points in the other direction.

availability	early (wk 40-52)	late (wk 1-20)
Oracle (collection-dated)	96.7	88.1
As-of, 8-16 wk window	16.5	95.6
As-of, 4-12 wk window	19.7	91.3

Table 2.2 — The same data split by calendar phase instead. Oracle early-minus-late is +8.6 points; as-of late-minus-early is +79.0 points.

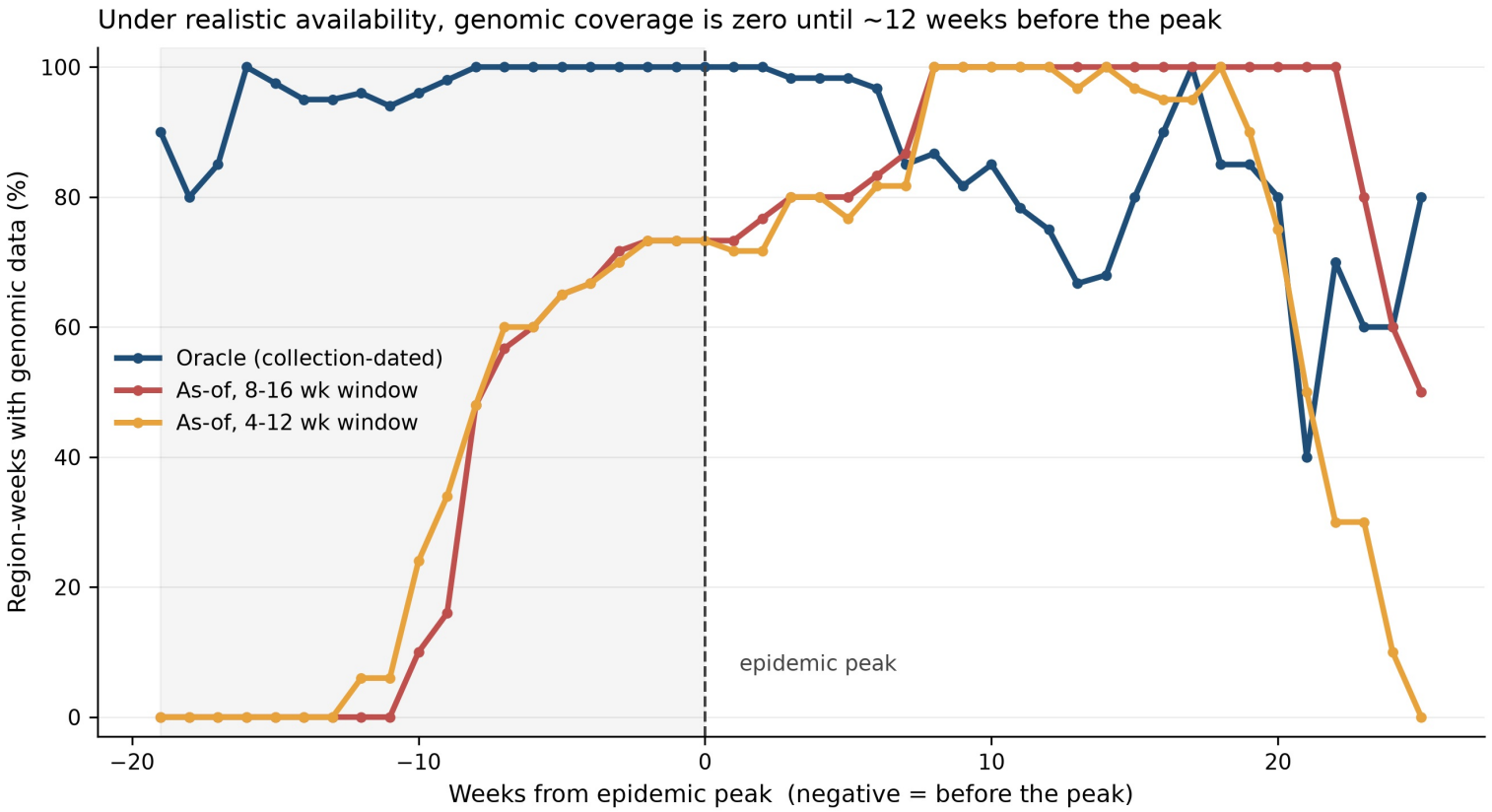


Figure 2.1 — Coverage against weeks from the epidemic peak. Under publication-dating, coverage is identically zero until roughly 12 weeks before the peak and only reaches the oracle level after it. The shaded region is the pre-peak half.

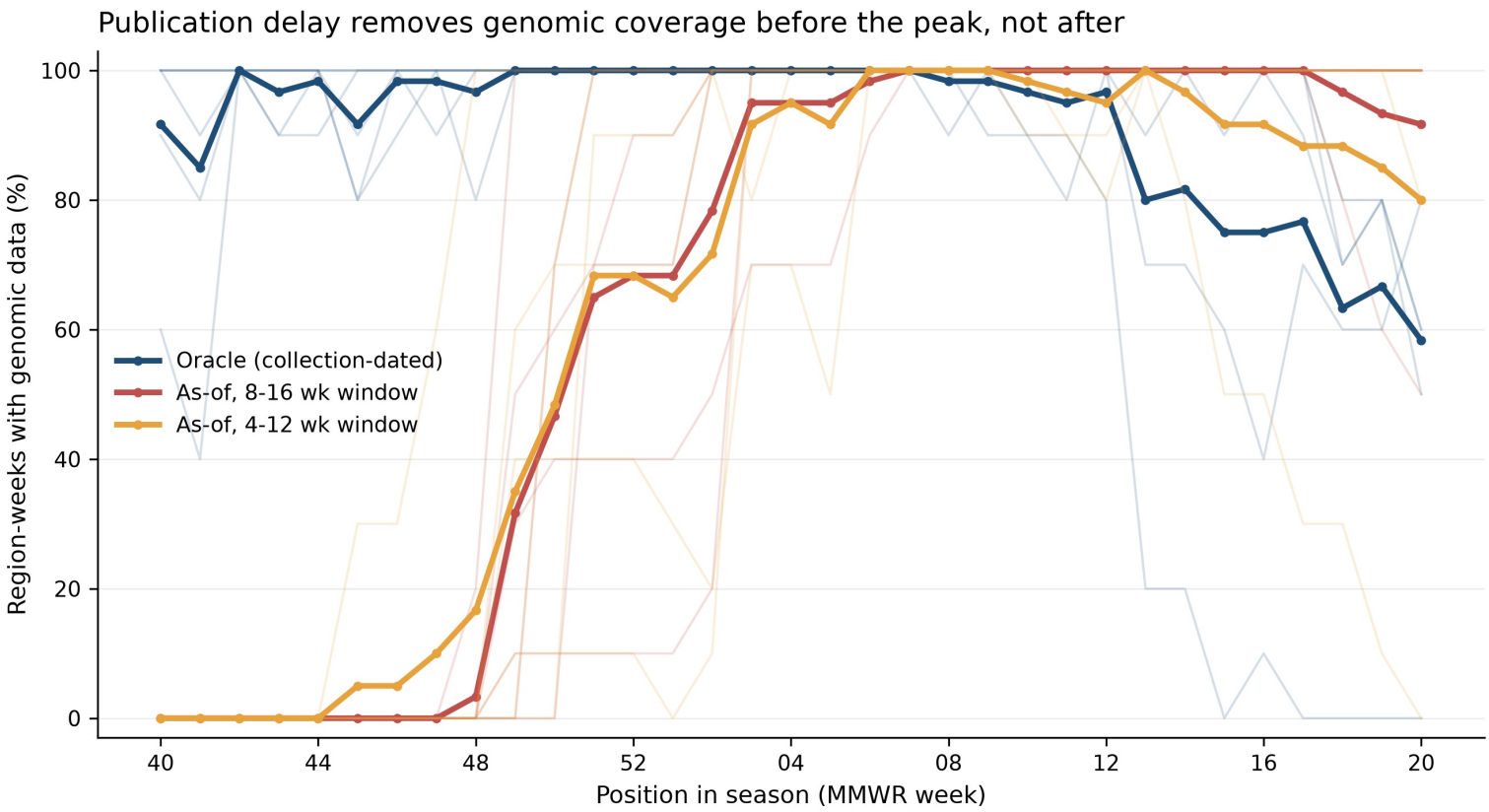


Figure 2.2 — The same quantity against position in the season. Faint lines are individual seasons. Publication delay removes coverage from the season's opening weeks, not from its tail.

season	consecutive onset weeks with zero coverage	MMWR range
2016-17	11	40–50
2017-18	8	40–47
2018-19	9	40–48
2019-20	10	40–49
2022-23	9	40–48
2023-24	9	40–48

Table 2.3 — Every season opens with a run of weeks in which no region has any usable genomic data, beginning at MMWR week 40 in all six.

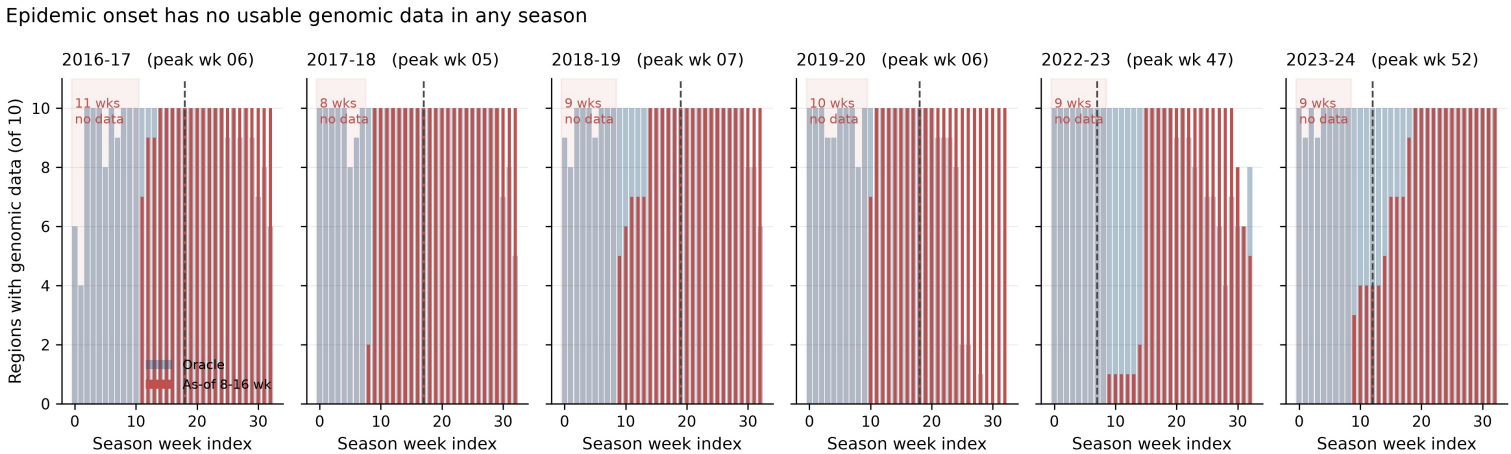


Figure 2.3 — Per season, the number of regions with genomic data each week. Blue is collection-dated, red is publication-dated. The shaded block marks the onset run with no usable data; the dashed line is the epidemic peak.

season	n_sequences	p25	median	p75	p90	median_weeks	pct_over_12wk
2016-17	3096	43	63	161	238	9	39
2017-18	4057	61	89	174	256	13	52
2018-19	5434	69	82	112	142	12	46
2019-20	2321	62	71	83	111	10	23
2022-23	5183	79	100	135	226	14	69
2023-24	12050	51	94	150	273	13	56
ALL	32141	62	86	134	237	12	52

Table 2.4 — Collection-to-submission delay in days, per season. The season medians span 63–100 days, so the delay should be quoted as a range rather than a point.

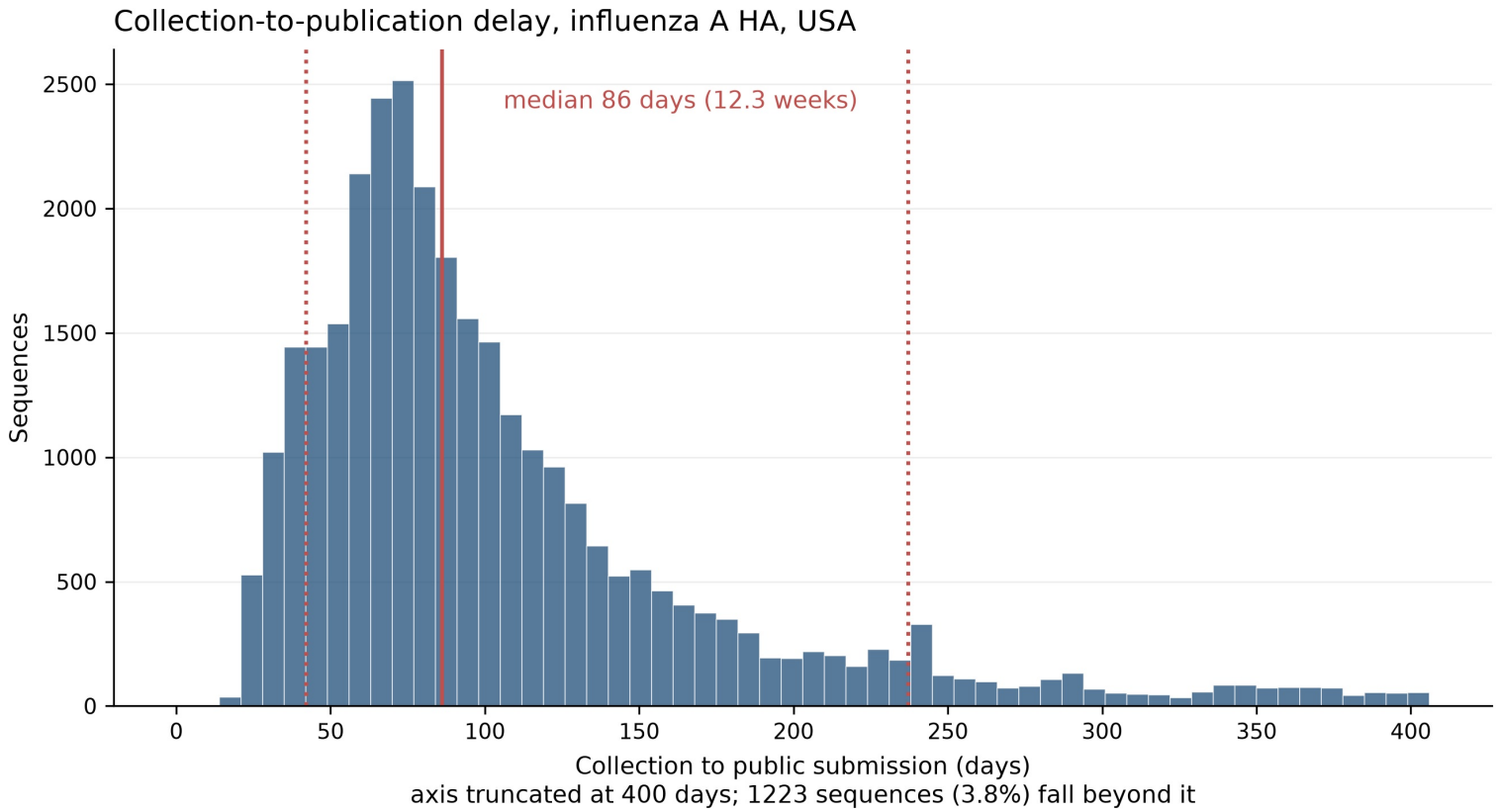


Figure 2.4 — Delay distribution across all 32,141 sequences. Median 86 days (12.3 weeks). The axis is truncated at 400 days; 3.8% of sequences fall beyond it.

Interpretation

The finding is not that delay reduces the amount of genomic data — it is *where* the reduction falls. Delay does not thin the data evenly. It removes it almost entirely from the onset weeks and returns it after the peak, by which time the epidemic's trajectory is largely determined. After the peak, publication-dated coverage is in fact marginally *higher* than collection-dated (−3.0 points), because the 8–16 week window aggregates several collection weeks.

> **Correction recorded.** Under an earlier, censored fetch window this pattern read as a *crossover* — oracle coverage falling late while as-of rose. Correcting the publication window showed most of that apparent oracle decline was an artifact of truncated late-season collections. The claim is one-sided loss, not relocation.

3. E2 — Baseline models

Objective

Establish what must be beaten before any genomic claim is meaningful, and determine which horizons support claims at all.

Methodology

Three reference models on the five rolling-origin folds. **B0 persistence** carries the last published wILI forward. **B1 climatological** uses the mean wILI for that (region, MMWR week) across strictly prior training seasons. **B2 gradient boosting** (`HistGradientBoostingRegressor`) uses eight as-of ILI lags, region and position in season. Metrics are MAE and RMSE; proper interval scores are deferred to a later phase.

Results

horizon	B0	B1	B2	B1 skill %	B2 skill %
1	0.768	1.179	0.688	-53.536	10.475
2	1.064	1.201	0.893	-12.875	16.036
3	1.335	1.221	1.060	8.523	20.551
4	1.593	1.238	1.171	22.266	26.528
6	2.051	1.260	1.369	38.579	33.257
8	2.408	1.256	1.443	47.832	40.089

Table 3.1 — Pooled MAE by horizon, with skill against persistence. Shown for orientation only; the per-fold tables below are the reportable view.

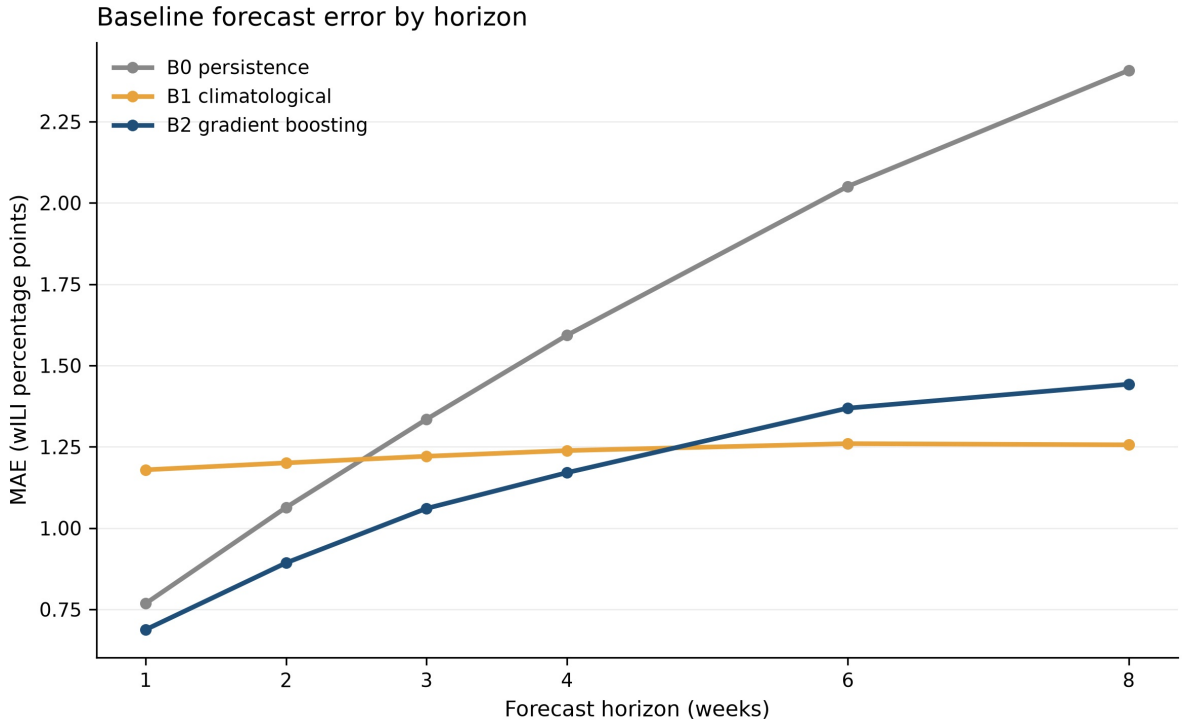


Figure 3.1 — Error by horizon. Persistence degrades steeply with horizon while climatology is flat; they cross between 2 and 3 weeks.

fold	1	2	3	4	6	8
fold1	-26.0	10.5	30.1	41.6	54.6	61.5
fold2	-44.8	-7.9	12.3	25.2	40.0	45.5
fold3	-11.6	13.5	25.5	34.2	46.0	52.7
fold4	-172.6	-93.8	-51.1	-24.6	3.5	19.4
fold5	-17.9	13.4	29.0	37.9	48.5	55.9

Table 3.2 — B1 climatological skill over persistence (%), per fold. Negative at h=1 in every fold; positive at h=6 and h=8 in every fold.

fold	1	2	3	4	6	8
fold1	17.5	37.1	47.0	53.5	60.1	63.5
fold2	14.7	20.1	26.8	32.4	41.6	40.3
fold3	27.6	35.6	41.1	47.8	54.6	58.2
fold4	**-29.1**	**-38.1**	**-42.5**	**-35.9**	**-35.8**	**-20.9**
fold5	18.3	22.2	30.1	34.9	43.9	49.9

Table 3.3 — B2 gradient boosting skill over persistence (%), per fold. Positive at every horizon in four of five folds; fold 4 reverses throughout.

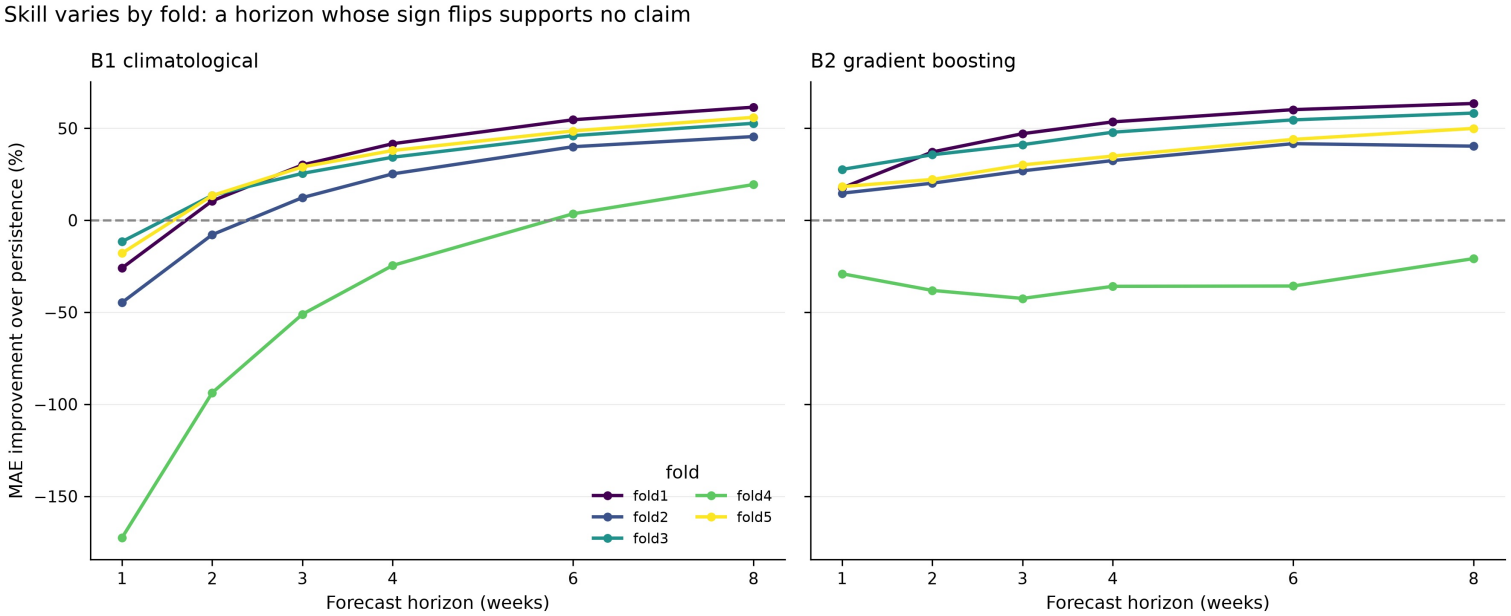


Figure 3.2 — Skill over persistence per fold. Fold 4 (light green) sits far below the others in both panels.

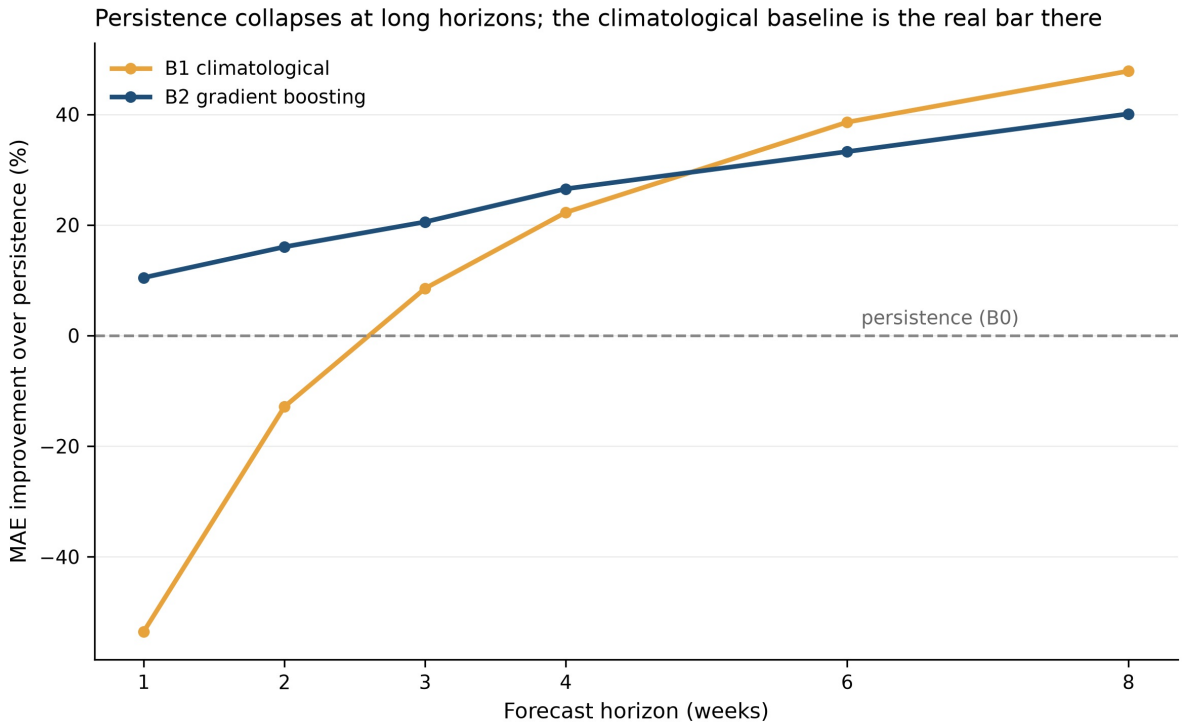


Figure 3.3 — Pooled skill over persistence. Beyond about 4 weeks, climatology rather than persistence is the honest bar.

Interpretation

Three claims survive every fold: persistence is unbeatable at one week; climatology beats persistence at six and eight weeks; and gradient boosting beats persistence at every horizon in four of the five folds. Horizons 2, 3 and 4 flip sign for climatology and support no claim.

Fold 4 is a finding, not noise. It trains only on pre-COVID seasons and tests on 2022-23, which peaked at MMWR week 47 against weeks 05–07 for every pre-COVID season — roughly ten weeks early. Every model that leans on seasonal shape fails there. This established the rule applied in all later experiments: pre-COVID folds and the gap-crossing fold are reported separately.

B2's seed standard deviation is exactly 0.0 at every horizon because `HistGradientBoostingRegressor` is deterministic given fixed training data. This is not evidence of stability, and the multi-seed protocol is reserved for the neural model in E6.

4. E3a — Genomic ablation on incidence forecasting

Objective

The project's central question. Does subtype composition add forecasting skill over case history, and how much of any gain is destroyed by publication delay?

Methodology

Eight arms, each adding a feature block to B2's epidemiological features. Everything else is held fixed. Arms A2–A4 use publication-dated composition (deployable); A5 and A6 use collection-dated composition (the literature-equivalent ceiling). A1 is a sampling-effort control, and A2s isolates what empirical-Bayes shrinkage contributes.

arm	feature block added	kind
A0	none — B2 exactly	reference
A1	sequencing effort only (GF3)	deployable
A2	as-of composition level (GF1, raw)	deployable
A2s	as-of composition level, shrunk	deployable
A3	as-of level + change (GF1+GF2)	deployable
A4	as-of, all families (GF1+GF2+GF4+GF5)	deployable
A5	oracle composition level	oracle
A6	oracle, all families	oracle

Table 4.1 — Ablation arms. The primary endpoint is A6−A4, the difference between the oracle and deployable full arms.

> **A design correction made before running.** The master table carries shrunk as-of composition but only raw oracle composition. Comparing "as-of" to "oracle" as originally specified would have compared *shrunk* against *raw* and confounded the availability contrast with the shrinkage. Both primary arms therefore use raw proportions, and A2s measures shrinkage separately.

Results

horizon	A0	A1	A2	A2s	A3	A4	A5	A6
1	0.688	0.674	0.682	0.681	0.676	0.684	0.684	0.697
2	0.893	0.894	0.894	0.879	0.868	0.875	0.902	0.916
3	1.060	1.085	1.054	1.044	1.031	1.052	1.074	1.058
4	1.171	1.186	1.173	1.157	1.169	1.168	1.190	1.189
6	1.369	1.368	1.370	1.356	1.364	1.392	1.378	1.378
8	1.443	1.443	1.452	1.440	1.440	1.428	1.489	1.446

Table 4.2 — MAE by arm and horizon (stratum = all, fold-averaged). Every arm sits within a few hundredths of A0.

contrast	meaning	h=1	h=2	h=3	h=4	h=6	h=8
A6-A4	**delay cost (PRIMARY)**	**FLIP**	**FLIP**	**FLIP**	**FLIP**	**FLIP**	**FLIP**
A4-A0	deployable genomics vs case history	**FLIP**	**FLIP**	**FLIP**	-8.2	-7.3	**FLIP**
A6-A0	oracle genomics vs case history	**FLIP**	**FLIP**	**FLIP**	-5.5	-5.7	**FLIP**
A5-A0	oracle level vs case history	**FLIP**	**FLIP**	**FLIP**	-5.2	-5.3	-7.9
A2-A0	as-of level vs case history	**FLIP**	**FLIP**	**FLIP**	-4.8	**FLIP**	**FLIP**
A1-A0	sequencing effort vs case history	**FLIP**	**FLIP**	**FLIP**	**FLIP**	**FLIP**	**FLIP**
A4-A1	genomics beyond effort	**FLIP**	**FLIP**	**FLIP**	**FLIP**	**FLIP**	**FLIP**
A2s-A2	what shrinkage buys	**FLIP**	**FLIP**	**FLIP**	**FLIP**	**FLIP**	**FLIP**

Table 4.3 — Sign consistency across the four pre-COVID folds. Not one arm produces a sign-consistent improvement at any horizon. Where results are consistent they are consistently negative.

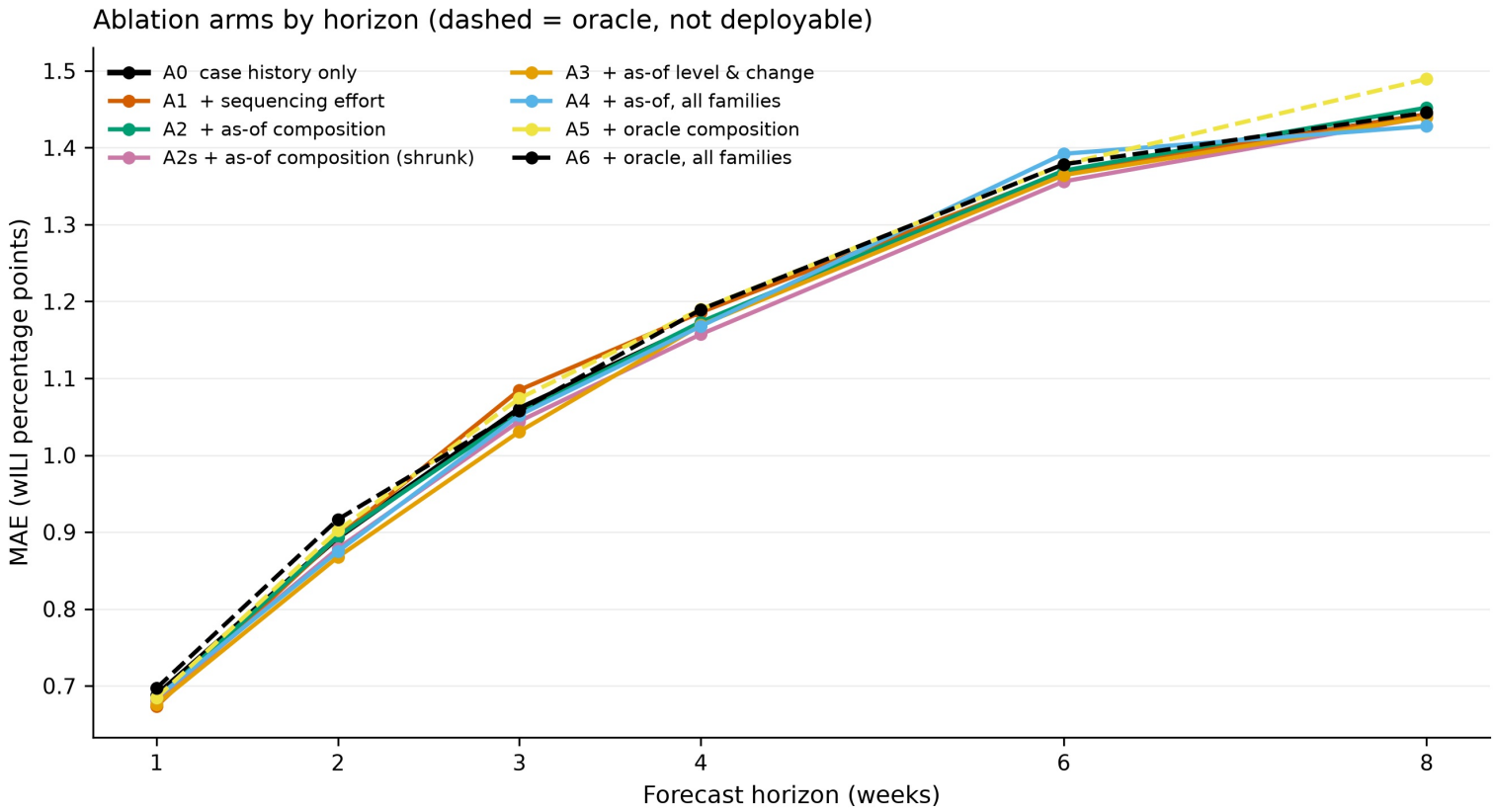


Figure 4.1 — All eight arms by horizon. The lines are nearly indistinguishable, which is the result.

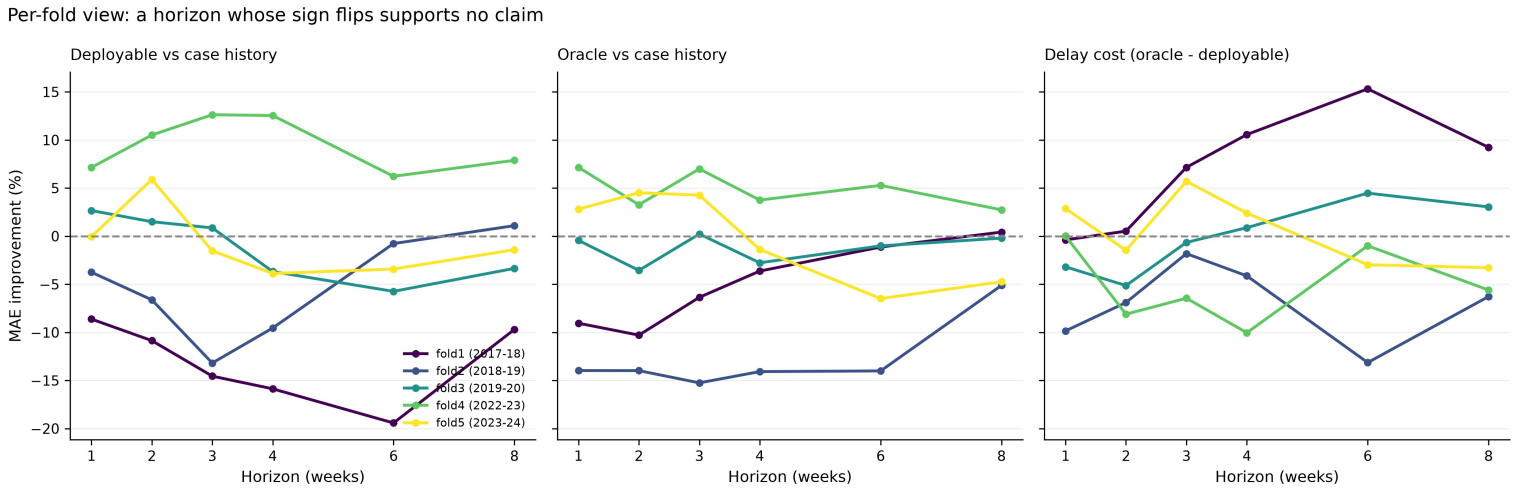


Figure 4.2 — Per-fold view of the three key contrasts. Folds disagree in sign in all three panels.

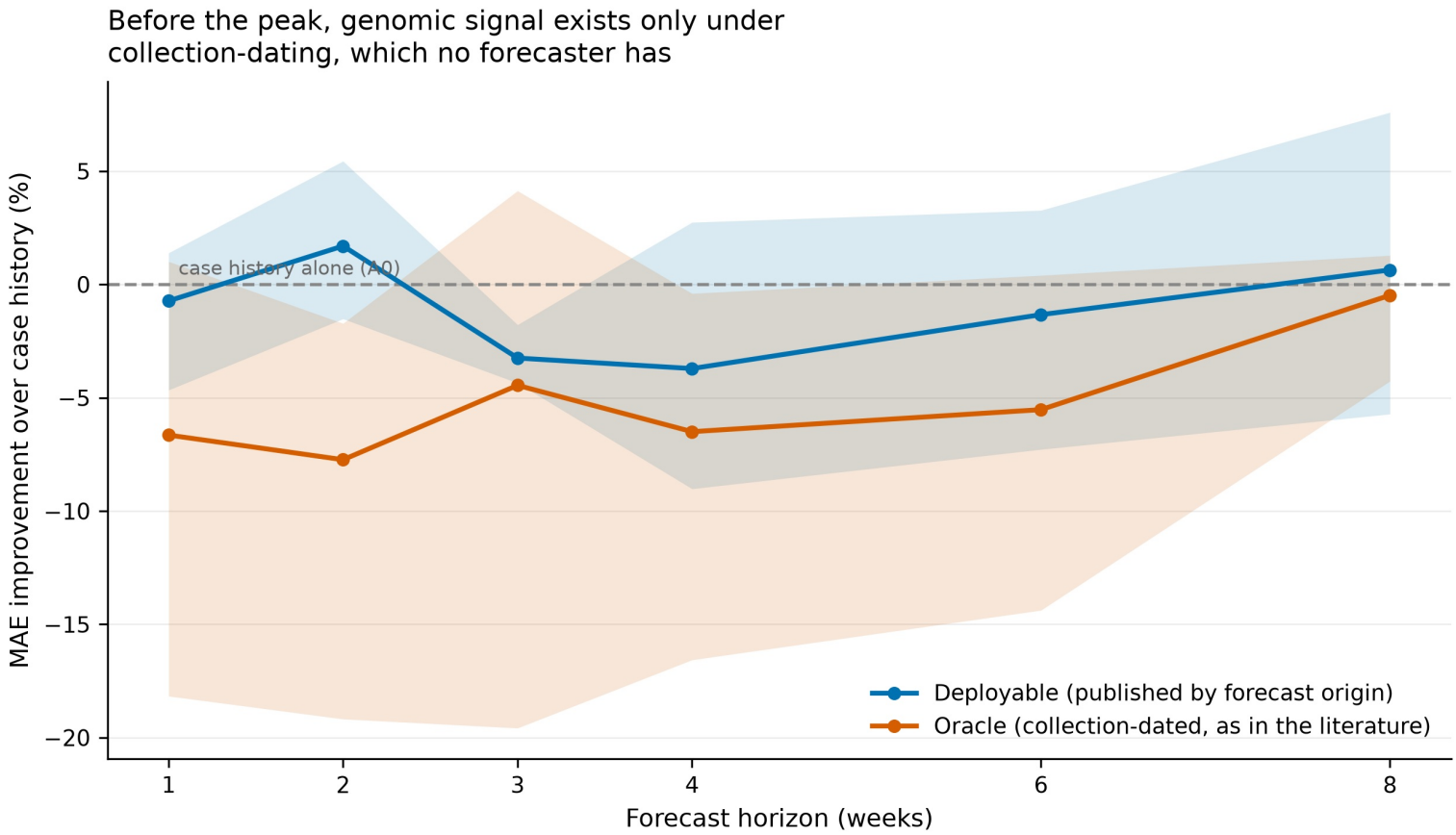


Figure 4.3 — Deployable and oracle genomic skill over case history, pre-peak. ***Note:*** *this figure's title was written before the null result was known and overstates it; both curves lie below zero, meaning neither availability condition helps. The title should be corrected before reuse.*

Controls — what makes a null credible

control	h=4	h=6
positive control (target + noise)	+50.2%	+52.9%
pure noise, 7 columns	FLIP	−2.5%
missingness indicator alone	FLIP	FLIP
A4 genomic, 7 columns	−8.2%	−7.3%

Table 4.4 — Controls run on every execution. The positive control reaches +37.6% to +56.0% across all folds and horizons, so the harness detects signal that is present. The missingness indicator does nothing, so the harm is not the NaN pattern.

horizon	A0 MAE (pre-peak)	bootstrap half-width	detectable effect (%)
1	0.631	0.0345	5.5
2	0.819	0.0553	6.8
3	1.012	0.0636	6.3
4	1.158	0.0832	7.2
6	1.408	0.1517	10.8
8	1.618	0.1340	8.3

Table 4.5 — Smallest delay cost this design could have resolved. 8 of 30 fold × horizon intervals exclude zero.

Interpretation

The primary endpoint — the delay cost — flips sign at all six horizons. There is no measurable cost because there is no signal to lose: the oracle arms do not help either. Where genomics is sign-consistent it is consistently *negative*, at a magnitude consistent with adding uninformative columns to a gradient booster on roughly 1,650 training rows.

This is a bounded result, not a universal one. It concerns **subtype composition** — H3N2 versus H1N1 — which is an extremely coarse genomic summary. Clade-level or antigenic-distance features cannot be tested here because the ingestion pipeline retained sequence metadata only and never stored nucleotides.

Fold 4 reverses sign (A4−A0 is +7.1% to +12.6%). That is a single fold in an opposite epidemiological regime and supports no claim either way; it is recorded so it is not later mistaken for a finding.

5. E3b — Subtype-dominance forecasting and the delay cost

Objective

E3a left the delay question unanswerable: with no genomic signal for incidence, there was nothing for delay to destroy. E3b changes the *target*, not the question. Composition is the natural predictor of composition, so this asks whether a region's eventual subtype mix is forecastable from what is known at the origin, and what delay costs that.

Methodology

The target is `oracle_prop_h3n2(r, t+h)` — the H3N2 share of sequences collected in region *r* during week *t*+*h*, the eventual truth about what circulated, exactly analogous to final revised wILI. Five carry-forward predictors are compared in a 2×2 of national versus regional and oracle versus as-of, plus a trivial rule.

> **Baseline discipline, learned the hard way.** A first pass reported 66% dominance accuracy for the oracle against 51% for the deployable predictor. That was very nearly an artifact: "always predict whichever subtype dominates *this* season" also scores 0.661–0.666. The honest trivial rule is **train-majority** — predict whichever subtype dominated the *training* seasons — which scores 0.506–0.513, because dominance flips hard between seasons (H3N2 led 94.5% of region-weeks in 2016-17 and 23.2% in 2019-20). All results below use that floor.

Results

predictor	1	2	3	4	6	8
nat ORACLE	**0.1803**	**0.1759**	**0.1794**	**0.1805**	**0.1923**	**0.2015**
reg ORACLE	0.2216	0.2115	0.2206	0.2244	0.2345	0.2451
nat as-of	0.3471	0.3555	0.3704	0.3796	0.4047	0.4182
reg as-of	0.3753	0.3801	0.3953	0.4052	0.4264	0.4355
train-majority	0.4920	0.4902	0.4930	0.4919	0.4916	0.4831

Table 5.1 — MAE on the H3N2 share (pre-COVID folds). Collection-dated national composition is roughly half the error of the publication-dated version.

predictor	1	2	3	4	6	8
nat ORACLE	**0.732**	**0.706**	**0.699**	**0.686**	**0.641**	**0.608**
reg ORACLE	0.655	0.663	0.672	0.652	0.629	0.608
nat as-of	0.469	0.465	0.442	0.433	0.403	0.393
reg as-of	0.472	0.474	0.446	0.435	0.428	0.429
train-majority	0.499	0.501	0.498	0.500	0.502	0.516

Table 5.2 — Binary dominance accuracy. Under realistic availability the predictor is at or below a coin flip at every horizon.

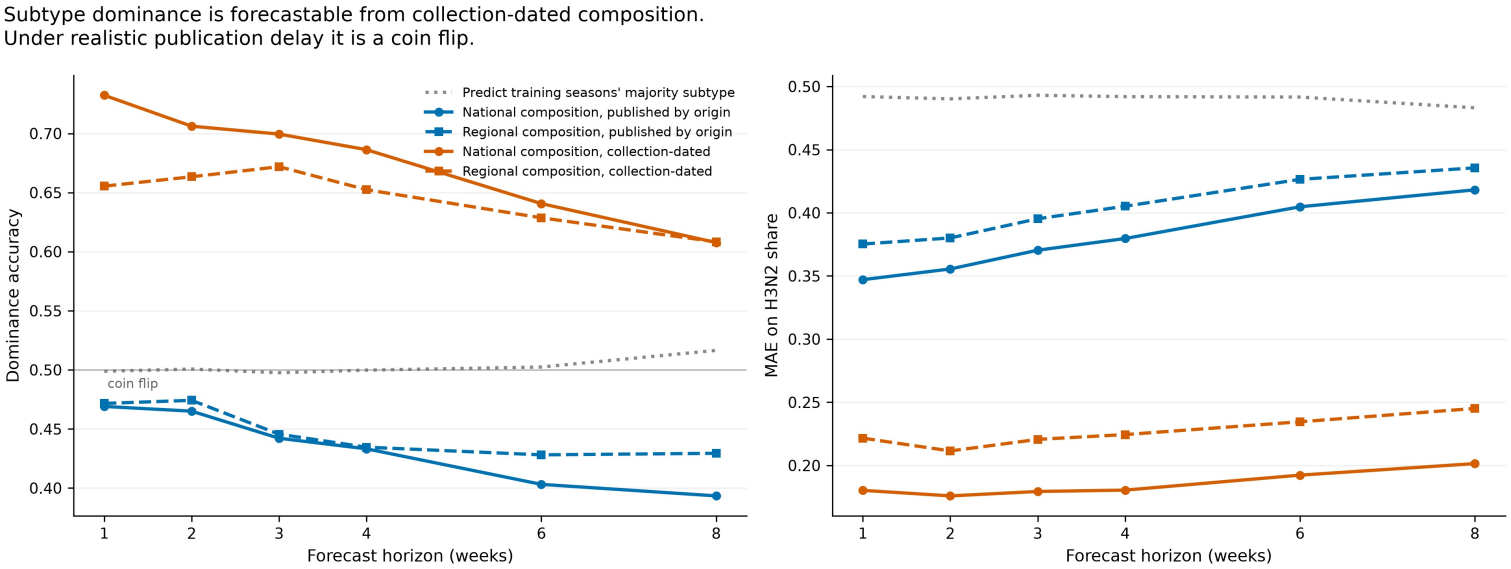


Figure 5.1 — Dominance accuracy (left) and MAE (right) by horizon. Orange is collection-dated, blue publication-dated, solid national and dashed regional. The grey dotted line is the trivial rule.

contrast	meaning	h=1	h=2	h=3	h=4	h=6	h=8
nat ORACLE vs nat as-of	**DELAY COST, national (PRIMARY)**	**+43.5**	**+45.4**	**+47.2**	**+48.8**	**+49.7**	**+49.8**
reg ORACLE vs reg as-of	delay cost, regional	+35.7	+38.3	+38.9	+39.7	+40.9	+40.4
reg as-of vs nat as-of	regional information, deployable	-8.4	-7.2	-7.0	-6.9	-5.5	-4.4
reg ORACLE vs nat ORACLE	regional information, oracle	-23.1	-19.7	-22.7	-24.1	-22.5	-22.3
nat as-of vs train-majority	deployable genomics vs trivial rule	+29.8	+27.9	+25.3	+23.2	+17.8	**FLIP**
nat ORACLE vs train-majority	oracle genomics vs trivial rule	+61.6	+62.2	+62.1	+62.0	+59.9	+57.3

Table 5.3 — Sign-consistent MAE skill across pre-COVID folds. The delay cost is positive and large at every horizon; regional information is consistently negative.

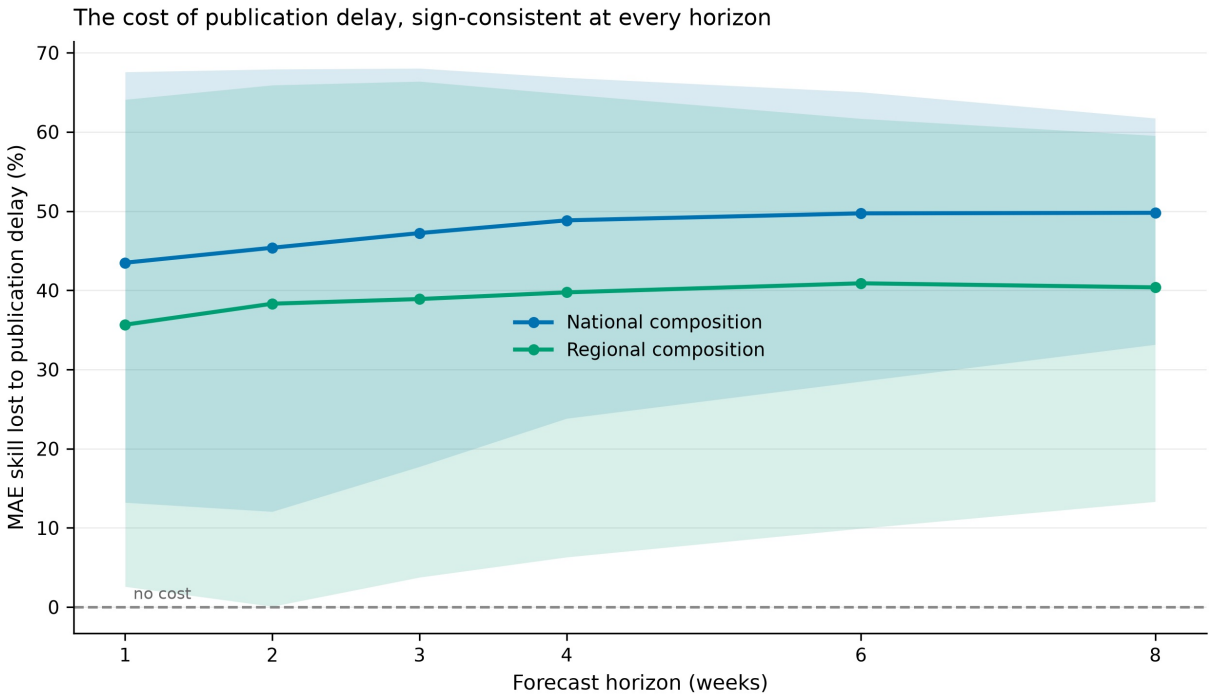


Figure 5.2 — The delay cost by horizon, with the per-fold range shaded. Both national and regional composition lose 35–50% of skill to publication delay.

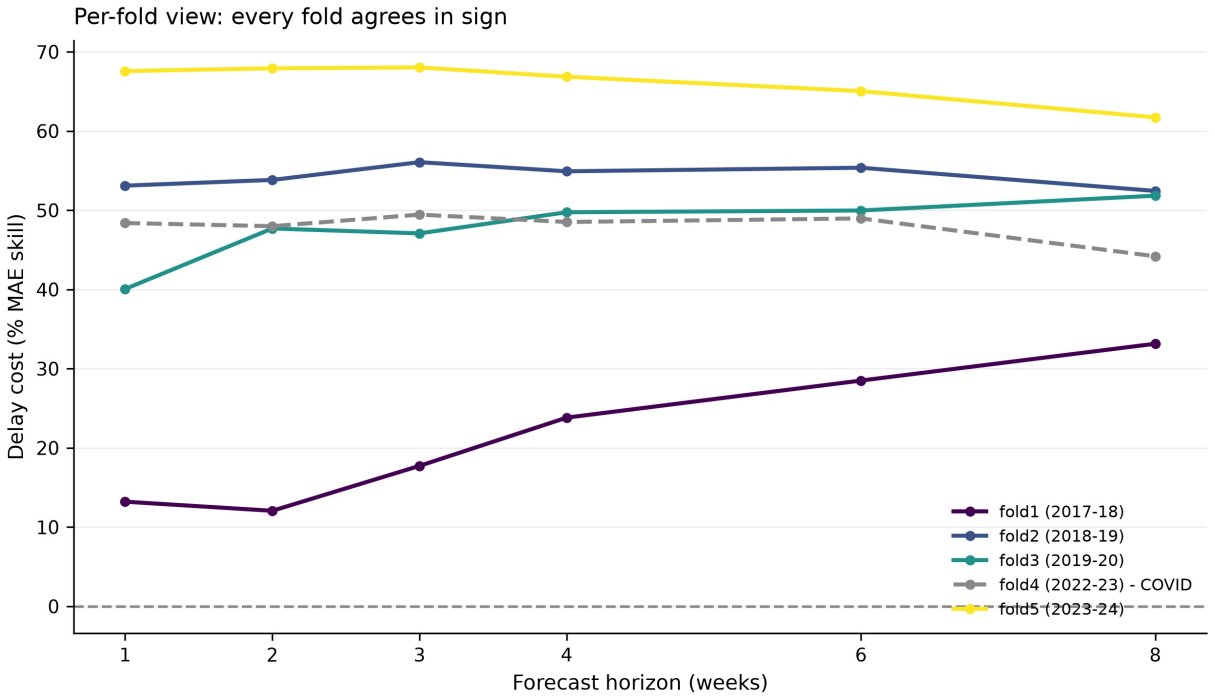


Figure 5.3 — Per-fold view of the primary endpoint. All five folds agree in sign, including the COVID fold (dashed).

stratum	1	2	3	4	6	8
pre-peak	**49.9**	**52.4**	**53.6**	**54.0**	**55.5**	**54.4**
peak or after	25.7	27.6	29.1	31.0	28.0	36.2
all	43.5	45.4	47.2	48.8	49.7	49.8

Table 5.4 — The delay cost by epidemic phase. It is markedly larger before the peak — independent corroboration of the E1 mechanism on a different target.

scope	1	2	3	4	6	8
all rows	26.3	24.1	25.7	25.3	23.8	21.4
both present	23.3	21.3	22.5	21.6	23.8	24.8

Table 5.5 — Availability or staleness? Rows where both variants carry real data isolate staleness alone. It accounts for most of the gap.

contrast	1	2	3	4	6	8
nat ORACLE vs nat as-of	48.4	48.0	49.5	48.5	49.0	44.2
nat ORACLE vs train-majority	63.2	60.9	60.8	59.0	58.0	52.0
nat as-of vs train-majority	28.7	24.8	22.4	20.3	17.6	14.0
reg ORACLE vs nat ORACLE	-38.2	-40.9	-37.9	-25.8	-21.5	-7.6
reg ORACLE vs reg as-of	34.1	30.6	33.7	37.5	39.4	41.0
reg as-of vs nat as-of	-8.3	-5.6	-5.1	-3.6	-2.4	-1.8

Table 5.6 — The COVID-discontinuity fold, reported separately. The delay cost holds there too.

Interpretation

Publication delay costs **43.5–49.8%** of subtype-dominance forecasting skill, sign-consistent at every horizon in every fold, with all 30 of 30 bootstrap intervals excluding zero. In accuracy terms, 73.2% falls to 46.9% at one week — from genuinely useful to worse than a coin flip.

The cost is staleness, not absence. On region-weeks where both variants carry real data, the accuracy gap is 21.3–24.8 points of the full 21.4–26.3, so roughly 85–95% of the damage is that the sequences one does have are too old — a sharper and more actionable claim than "data is missing".

Regional sequencing adds nothing. Regional composition is consistently *worse* than national at every horizon, by 4.4–8.4% when publication-dated and 19.7–24.1% when collection-dated. Regional sequence counts are small and aggregation wins. This is a clean negative with direct surveillance-design relevance.

No model helps. Gradient boosting on the same predictors does not beat carry-forward at any horizon. Subtype-dominance forecasting is an information-*availability* problem, not a modelling problem, and the result is reported in its model-free form because that is both the honest and the stronger version.

A nuance worth stating. Publication-dated composition beats the trivial rule on MAE (+17.8% to +29.8%) while sitting *below* it on accuracy. Delayed data gives a reasonable estimate of the *mix* but cannot call the *winner*, because its estimate hovers near 0.5 and thresholding there is a coin flip.

6. E3c — What would it take to recover the signal?

Objective

E3b is a diagnosis. E3c asks the question a surveillance programme would ask next: what would have to change for the signal to survive?

Methodology

A counterfactual simulation on real submission dates. Two levers are varied jointly: submission shifted *D* days earlier, and the collection window used at the forecast origin. Current practice is the 8–16 week window at *D*=0, and every cell is scored against it. *D* is a **shift, not a resulting delay**: the median delay is 86 days, so *D*=56 corresponds to a median of about 30 days.

Results

window	0	14	28	56	84
L0-2	-32.8	-28.3	-16.3	+29.8	+36.2
L0-4	**-29.2**	**-17.6**	**-4.0**	**+33.3**	**+36.7**
L2-6	-19.4	-7.0	+15.4	+27.4	+30.4
L4-12	+1.9	+9.5	+16.3	+20.2	+22.6
L8-16	**+0.0**	**+2.3**	**+3.4**	**+5.2**	**+5.6**

Table 6.1 — MAE skill against current practice (%), pre-COVID folds, averaged over horizons. Columns are the submission shift in days; rows are the collection window.

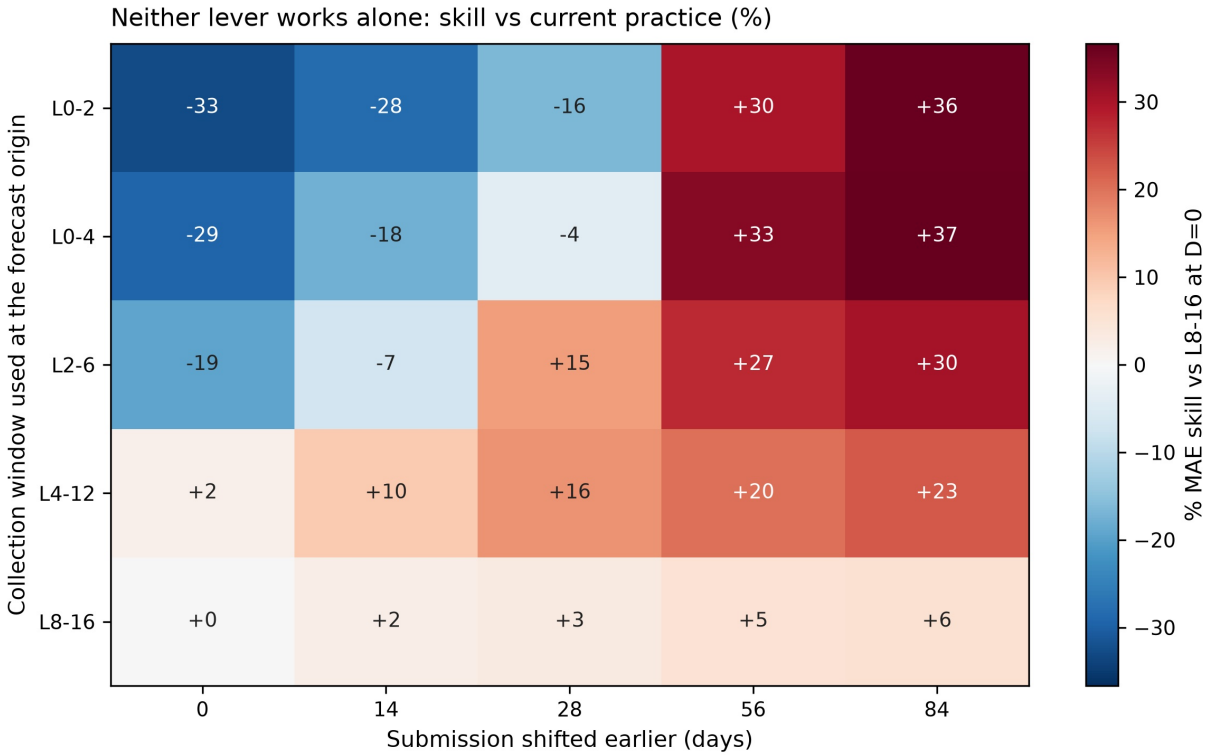


Figure 6.1 — The joint surface. Blue is worse than current practice, red better. The bottom row is faster submission alone; the left column is a tighter window alone. Both are weak or harmful; the top-right corner is where they combine.

window	0	14	28	56	84
L0-2	99	89	69	6	4
L0-4	89	70	41	5	4
L2-6	72	46	19	11	11
L4-12	32	26	20	19	19
L8-16	36	34	34	34	34

Table 6.2 — The mechanism: percentage of forecast origins with no usable sequence. A tight window at today's submission speed is almost always empty.

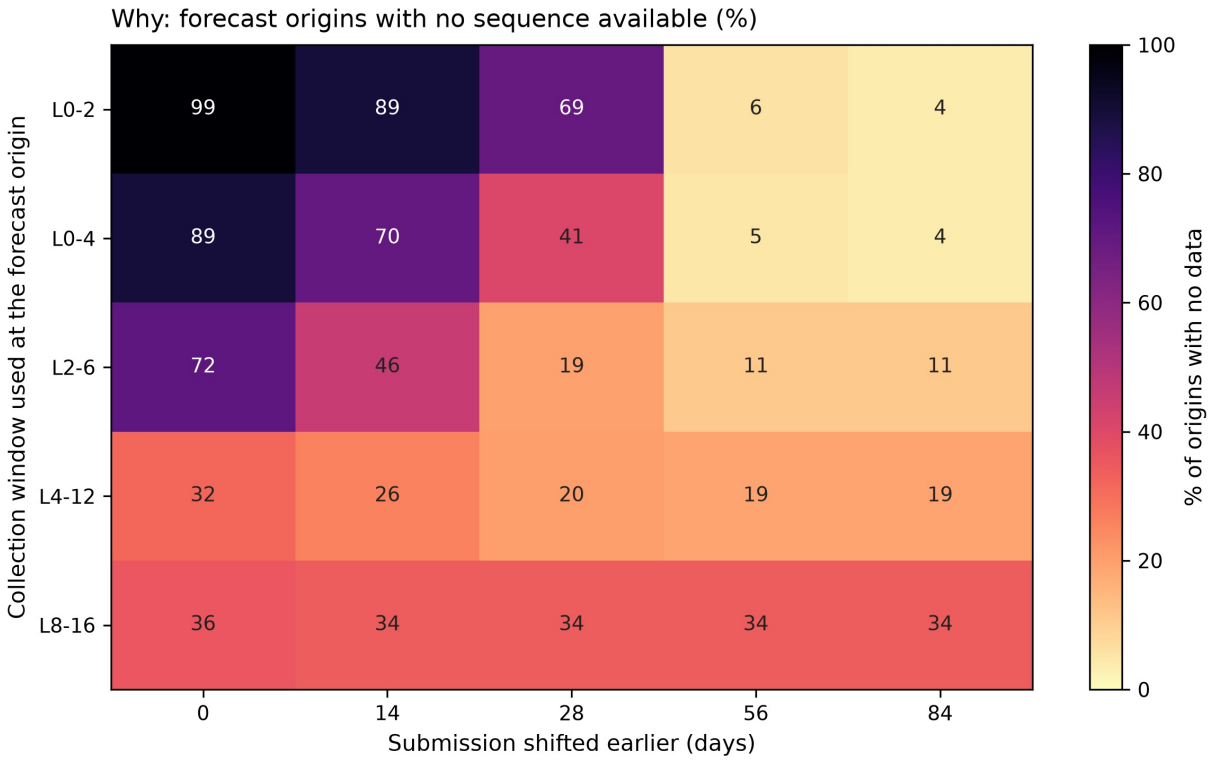


Figure 6.2 — Data availability across the same grid. Faster submission does not add signal; it makes the tight window usable, and the tight window is what carries the signal.

window	0	14	28	56	84
L0-2	0.502	0.498	0.508	0.596	0.581
L0-4	0.496	0.500	0.507	0.582	0.589
L2-6	0.492	0.492	0.506	0.543	0.567
L4-12	0.435	0.461	0.491	0.496	0.519
L8-16	0.434	0.436	0.435	0.442	0.442

Table 6.3 — Dominance accuracy across the surface. Current practice sits at 0.434; the best joint cell reaches 0.589.

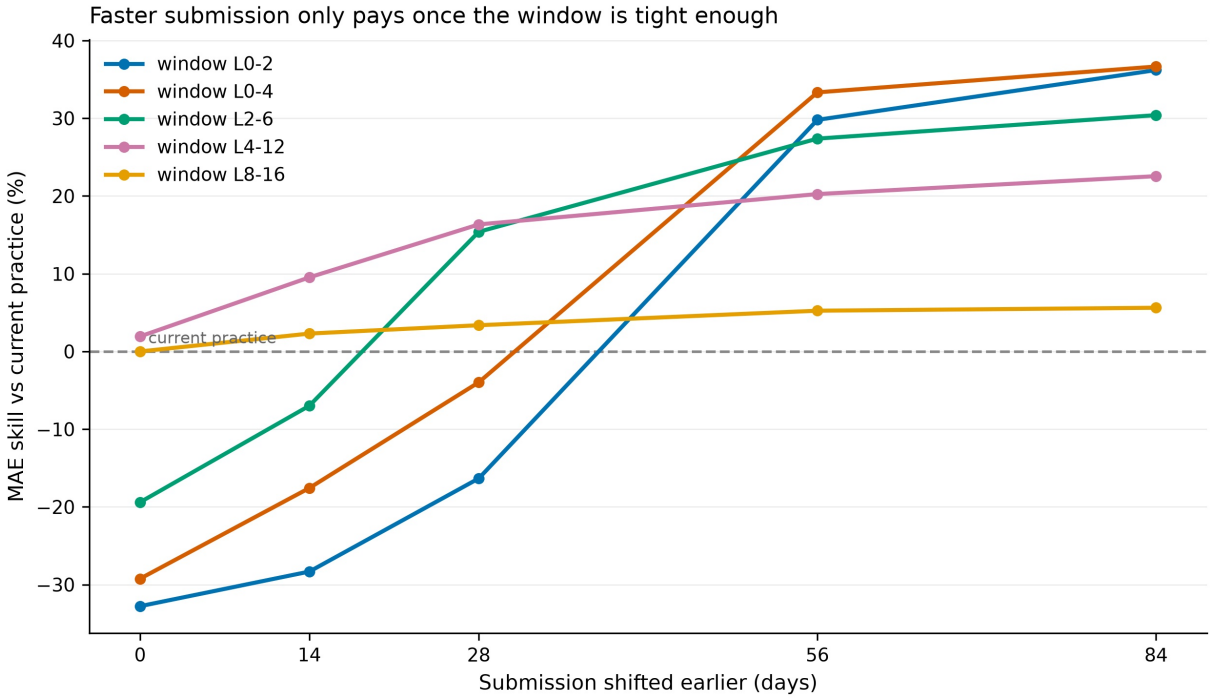


Figure 6.3 — Skill against submission speed, one line per window. The 8–16 week window (yellow) is almost flat: speeding up submission without tightening the window buys almost nothing.

intervention	cell	fold1	fold2	fold3	fold4	fold5
faster submission ALONE	L8-16, D=84	+1.3	+7.5	+2.8	+16.6	+11.1
tighter window ALONE	L0-4, D=0	-55.9	-10.5	-42.2	-25.5	-8.3
BOTH	**L0-4, D=56**	**+11.5**	**+37.2**	**+39.3**	**+40.1**	**+45.9**

Table 6.4 — The three headline cells at h=4, per fold. The single-lever interventions are weak or reliably harmful; the joint one is positive in every fold.

window	shift (days)	h=1	h=2	h=3	h=4	h=6	h=8
L0-2	0	-44.7	-41.6	-36.1	-33.1	-23.8	FLIP
L0-2	14	-39.4	-36.4	-31.3	-28.5	FLIP	FLIP
L0-2	28	FLIP	FLIP	FLIP	FLIP	FLIP	FLIP
L0-2	56	FLIP	FLIP	FLIP	+30.7	+31.2	+31.0
L0-2	84	FLIP	FLIP	+36.4	+36.5	+37.4	+38.1
L0-4	0	-39.9	-37.9	-32.0	-29.2	-21.3	FLIP
L0-4	14	FLIP	FLIP	FLIP	FLIP	FLIP	FLIP
L0-4	28	FLIP	FLIP	FLIP	FLIP	FLIP	FLIP
L0-4	56	FLIP	+33.3	+33.6	+33.5	+34.0	+34.1
L0-4	84	+34.7	+35.9	+36.5	+36.8	+37.7	+38.2
L2-6	0	-27.9	FLIP	FLIP	FLIP	FLIP	FLIP
L2-6	14	FLIP	FLIP	FLIP	FLIP	FLIP	FLIP
L2-6	28	FLIP	FLIP	+15.3	+14.9	+16.9	+16.5
L2-6	56	+26.7	+26.9	+27.7	+27.3	+27.7	+28.0
L2-6	84	+29.4	+29.8	+30.6	+30.4	+31.0	+31.2
L4-12	0	+2.1	+1.6	FLIP	FLIP	FLIP	FLIP
L4-12	14	+9.3	+8.9	+9.6	+9.7	+10.5	+9.1
L4-12	28	+16.5	+15.7	+16.3	+16.5	+17.1	+16.1
L4-12	56	+20.7	+20.0	+20.0	+20.2	+20.7	+19.9
L4-12	84	+23.1	+22.4	+22.5	+22.6	+22.9	+21.9
L8-16	14	+2.5	+2.4	+2.5	+2.4	+2.1	+1.9
L8-16	28	+3.6	+3.5	+3.5	+3.5	+3.3	+2.8
L8-16	56	+5.9	+5.5	+5.3	+5.3	+4.9	+4.5
L8-16	84	+6.5	+5.9	+5.8	+5.7	+5.1	+4.6

Table 6.5 — Sign consistency for every cell against current practice. The corners are clean; the middle of the surface is a transition zone where the two levers trade off and no claim is available.

Interpretation

The result is an **interaction effect**, and it is the least obvious finding in the project. Faster submission with today's window recovers 5.6%. A tighter window at today's submission speed is 29.2% *worse* than doing nothing — sign-consistently negative — because 89% of forecast origins have no sequence in a 0–4 week window at all and the forecast falls back to a prior. Applied together they recover 33–37%.

The policy reading is therefore not "sequence faster" but "sequence faster *and then use the fresher data*". A programme that accelerates submission while continuing to read an 8–16 week window captures almost none of the benefit.

> **Stated limitation.** This is a counterfactual on real submission dates, not a forecast of what a faster system would achieve. It assumes *uniform* delay reduction — every sequence *D* days earlier — which no real intervention delivers; real speed-ups change the shape of the delay distribution, not only its location. The surface is an order-of-magnitude guide to which lever matters, not a service-level target.

7. E6 — The spatio-temporal graph model

Objective

The architecture plan specifies a graph over the ten HHS regions and mandates a density-matched random-graph control. This experiment answers the plan's third claim: which inter-region relation carries forecasting information?

Methodology — three layers

Layer A, stability (no model). Measures directly whether the inter-region correlation structure persists between seasons.

Layer B, feature screen (gradient boosting). Adds neighbour wILI lags to B2 under three neighbour definitions, with neighbour sets chosen from training weeks only and a random arm of identical density.

Layer C, the GNN. A per-region GRU over the lag window, one graph-convolution layer, and a multi-horizon head, trained under four adjacencies with architecture, parameter count, optimiser and seeds held fixed. Three seeds per fold.

> **Why Layer C is credible.** A first implementation scored 1.10 MAE at h=1 where plain persistence achieves 0.76. It had never learned to copy the most recent observation, so its null told us nothing about graphs — only that the model failed to train. Reporting that as "graphs do not help" would have been wrong. The model reported here predicts the **residual from persistence**, so a zero output already equals persistence and the network can only improve on it. It then beats persistence substantially, and only then is its indifference to the graph informative.

Results — Layer A

measure	value
Mean pairwise |r| between regions, within a season	0.850
Correlation of the correlation *matrix*, between season pairs	+0.100
Top-3 neighbour overlap between season pairs	38.0%
Train-chosen neighbours that are the test season's true top-3	34.7%
Chance rate for the above (3 of 9)	33.3%

Table 7.1 — Stability of the correlation structure. Regions move together strongly, but which regions pair with which carries almost no information across seasons.

fold	test season	chosen neighbours that are correct
fold1	2017-18	33.3%
fold2	2018-19	30.0%
fold3	2019-20	46.7%
fold4	2022-23	26.7%
fold5	2023-24	36.7%

Table 7.2 — Per fold, the overlap between neighbours selected on training seasons and the test season's actual top-3. The mean is indistinguishable from chance.

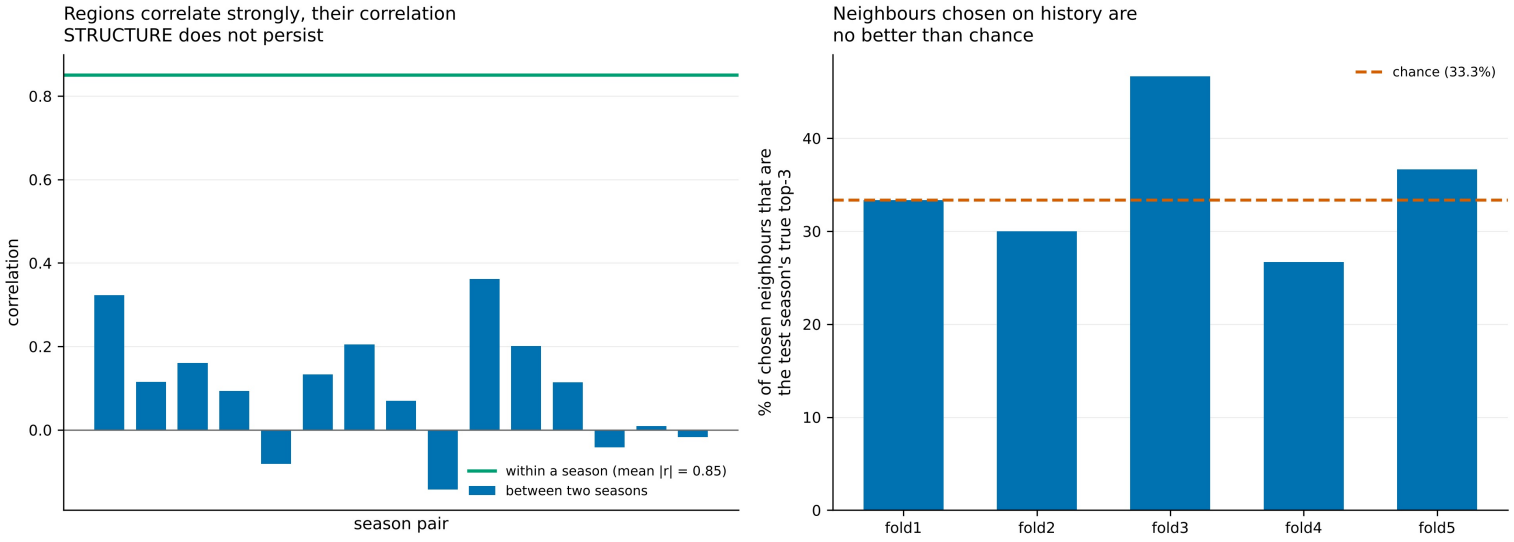


Figure 7.1 — Left: within-season correlation (green line) against between-season persistence of the correlation matrix (bars). Right: per fold, how often a historically-chosen neighbour is actually a top neighbour in the test season, against the chance rate.

Results — Layer B

arm	1	2	3	4	6	8
+ 3 RANDOM neighbours (control)	0.6170	0.7617	0.8308	0.8988	1.0717	1.0589
+ 3 most-correlated neighbours	0.6279	0.7749	0.8787	0.9619	1.1056	1.1592
+ all other regions (mean)	0.6350	0.7771	0.8783	0.9383	1.1006	1.1947
B2 own history only	0.6063	0.7306	0.8105	0.8679	0.9706	1.0839

Table 7.3 — MAE with neighbour features added to B2 (pre-COVID folds).

neighbour features added	h=1	h=2	h=3	h=4	h=6	h=8
+ 3 most-correlated neighbours	-3.9	-6.5	-9.2	-11.5	-14.5	-7.0
+ 3 RANDOM neighbours (control)	**FLIP**	-3.9	**FLIP**	**FLIP**	-11.3	**FLIP**
+ all other regions (mean)	-4.8	-6.8	-9.1	-8.7	-14.1	**FLIP**

Table 7.4 — Skill against own-history-only. Every neighbour definition hurts, and correlation-chosen neighbours hurt more than randomly-chosen ones at identical column counts.

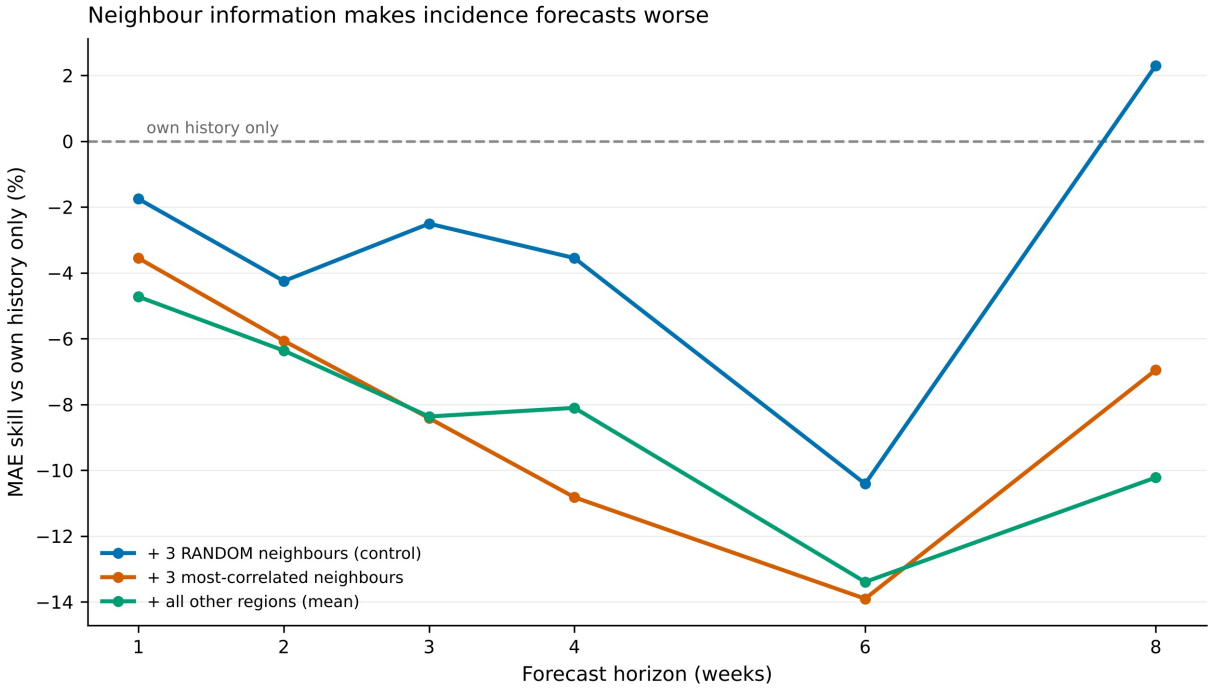


Figure 7.2 — Neighbour information degrades incidence forecasts. The random-neighbour control is the least harmful of the three.

Results — Layer C

model	1	2	3	4	6	8
B0 persistence	0.7640	1.0503	1.3070	1.5536	2.0183	2.4146
B2 boosting	0.6063	0.7306	0.8105	0.8679	0.9706	1.0839
GNN, no graph	**0.7067**	**0.8606**	**0.9683**	**1.0215**	**1.1258**	**1.1874**
GNN, correlation graph	0.6998	0.8793	0.9729	1.0533	1.1921	1.2912
GNN, fully connected	0.7172	0.8821	0.9880	1.0713	1.1943	1.2912
GNN, random graph	0.6988	0.8654	0.9753	1.0372	1.1665	1.2441

Table 7.5 — MAE by model and horizon (pre-COVID folds, mean over three seeds).

comparison	question	h=1	h=2	h=3	h=4	h=6	h=8
GNN no-graph vs B0 persistence	does the model learn?	**FLIP**	+18.4	+26.0	+34.6	+43.9	+49.8
GNN no-graph vs B2 boosting	stated limitation	-15.7	**FLIP**	**FLIP**	**FLIP**	**FLIP**	**FLIP**
GNN correlation vs no-graph	does the relation help?	**FLIP**	**FLIP**	**FLIP**	**FLIP**	**FLIP**	**FLIP**
GNN fully-connected vs no-graph		**FLIP**	**FLIP**	**FLIP**	**FLIP**	**FLIP**	-8.2
GNN random vs no-graph		**FLIP**	**FLIP**	**FLIP**	**FLIP**	**FLIP**	-4.9
GNN correlation vs RANDOM	**the G5 control**	**FLIP**	**FLIP**	**FLIP**	**FLIP**	**FLIP**	**FLIP**

Table 7.6 — Sign-consistent skill. The model learns; the graph does not help; and the correlation graph does not beat the density-matched random control.

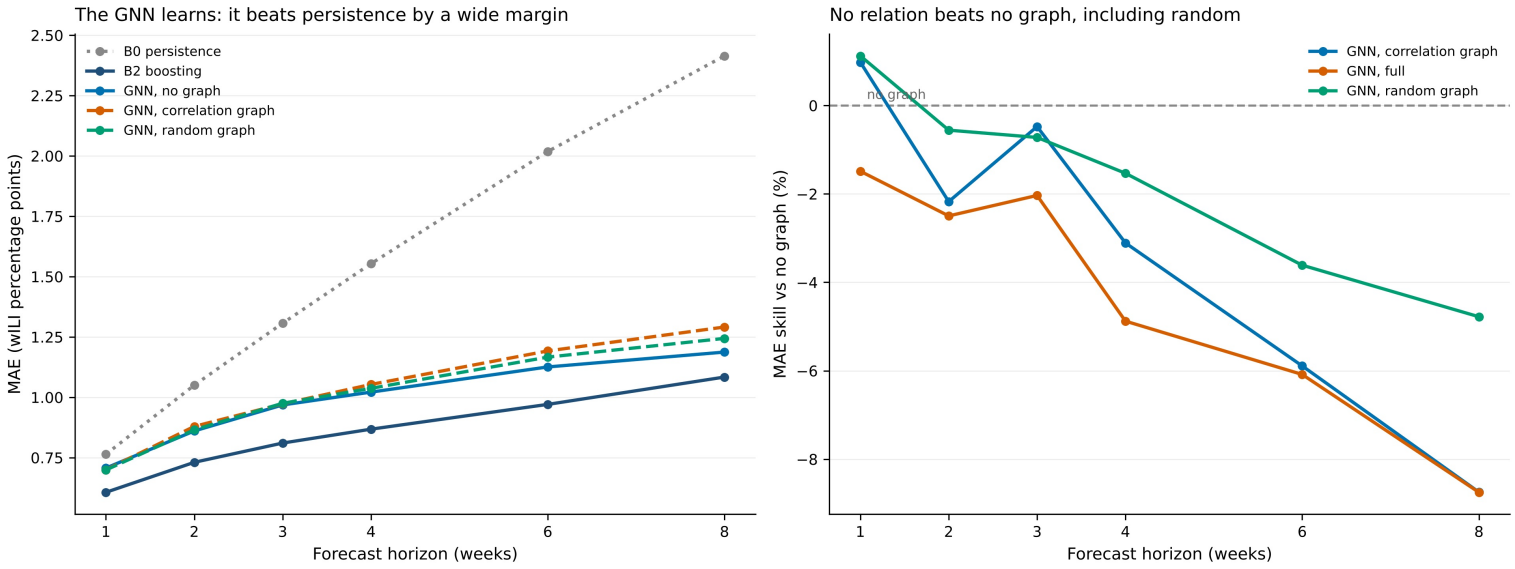


Figure 7.3 — Left: the GNN against persistence and gradient boosting. Right: each relation against the no-graph model — all within noise of zero.

Interpretation

The three layers agree, and the mechanism explains the behaviour. Within any season the ten regions are strongly correlated (mean |r| = 0.85), so constructing adjacency from correlation looks obviously justified. But the correlation *matrix* barely persists between seasons (matrix-to-matrix r = +0.10), and neighbours selected on training seasons are the test season's true top neighbours only 34.7% of the time against a 33.3% chance rate. **A correlation graph fitted on history is, for the next season, statistically a random graph** — and both the feature screen and the trained GNN behave exactly as that predicts.

The GNN itself works: it beats persistence by +18.4% to +49.8% at horizons 2 through 8, sign-consistent across folds. With that working instrument, no relation beats no-graph, and adding any graph is mildly harmful at the longest horizon — the signature of extra parameters fitting nothing.

> **Stated limitation.** The GNN does not beat gradient boosting (−15.8% at h=1, indistinguishable elsewhere). With ten nodes and roughly 1,650 training rows this is expected, and it is reported rather than hidden. The contribution here is the relation ablation and its mechanism, not the model's accuracy.

8. Consolidated comparison and progression

experiment	question	design	finding	outcome
E1	Where does delay remove genomic coverage?	model-free coverage accounting	Coverage vanishes before the peak, not after: 97.1% vs 34.8%, 8–11 dead onset weeks per season	**Positive**
E2	What must a genomic model beat?	B0 / B1 / B2 on 5 rolling folds	Three fold-invariant claims; fold 4 reverses and is a regime-change finding	**Positive**
E3a	Does composition forecast incidence?	8-arm ablation, 3 controls, power floor	No, at any horizon, oracle or as-of; detectable floor 5.5–10.8%	**Null**
E3b	Does composition forecast subtype dominance?	2×2 of national/regional × oracle/as-of	Yes; delay costs 43.5–49.8%, ~85–95% of it staleness; regional adds nothing	**Positive**
E3c	What would recover the lost signal?	counterfactual window × submission-speed surface	Neither lever alone; jointly +33 to +37%	**Positive**
E6	Which graph relation carries information?	stability + feature screen + GNN with G5 control	None; correlation graphs are statistically random across seasons	**Null**

Table 8.1 — The six experiments. Two produced nulls; both nulls are controlled and both are load-bearing for the positive results.

How the experiments depend on one another

The sequence is not six independent studies. Each one is the direct consequence of the previous result:

- **E1 → E3a.** E1 showed genomic data is missing exactly when a forecast is needed. E3a asked whether having it would have helped.
- **E3a → E3b.** E3a's null made the delay question unanswerable — no signal, nothing to lose. E3b changed the target to one where composition is the natural predictor, which made the delay cost measurable again.
- **E3b → E3c.** Having measured the damage, E3c asked what would repair it, and found the two obvious remedies fail independently.
- **E2 → E6.** E2 established the baselines and the per-fold reporting rule that E6's relation ablation is judged by.
- **E1 ↔ E3b.** The two corroborate each other without being fitted to each other: E1 measured the mechanism in coverage, E3b measured its consequence on a different target, and the delay cost is larger pre-peak (49.9–55.5%) than post-peak (25.7–36.2%) exactly as E1 predicts.

The role of the graph model

E6 is the architectural centrepiece of the original plan, and its result is negative. That is worth stating plainly: the spatio-temporal GNN does not improve regional influenza forecasting on this dataset, and no inter-region relation tested carries information beyond a random graph of the same density. What makes the experiment valuable is not the model but the *control and the mechanism* — the density-matched random graph was specified in advance, and the stability analysis explains why the relation fails rather than merely reporting that it does.

9. Novelty and research contributions

Contributions are separated into those demonstrated by the experiments above and those the work only gestures at. Claims are stated at the strength the evidence supports and no further.

9.1 Demonstrated contributions

C1. A delay-aware evaluation protocol for genomic forecasting features

What it is. A paired oracle-versus-as-of evaluation harness in which the same model, rows and folds are scored under collection-dating and publication-dating, with a positive control and a stated detectable-effect floor.

What exists already. Vintage-correct or "as-of" evaluation is standard for case-count nowcasting. Applying it to the *genomic* inputs of a forecasting pipeline, and reporting the oracle and deployable conditions side by side as the primary endpoint, is less common.

Evidence. E3a and E3b, with the controls in Table 4.4 and the power analysis in Table 4.5.

Limit. Demonstrated on one pathogen, one country and six seasons.

C2. Quantification of the delay cost, and its decomposition into staleness and absence

What it is. Publication delay costs 43.5–49.8% of subtype-dominance forecasting skill, and 85–95% of that is because available sequences are stale rather than because they are missing.

Why the decomposition matters. "Data is missing" and "the data you have is too old" imply different interventions. The decomposition is what makes E3c's design possible at all.

Evidence. Tables 5.3 and 5.5; all 30 bootstrap intervals exclude zero; the effect holds in the COVID fold.

Limit. Measured on subtype composition, the coarsest genomic resolution available in this dataset.

C3. The interaction between submission speed and collection window

What it is. Neither faster submission nor a tighter collection window recovers the lost signal alone — the second is reliably *harmful* — while jointly they recover 33–37%.

Why it is distinctive. It is a non-obvious interaction with a measured mechanism (Table 6.2), and it reverses the intuitive policy recommendation. It is the clearest actionable result in the project.

Evidence. Tables 6.1, 6.2 and 6.4; sign consistency in Table 6.5.

Limit. A counterfactual assuming uniform delay reduction.

C4. Correlation-based epidemic graphs fail a density-matched random control

What it is. Three independent layers — a stability analysis, a feature screen and a trained GNN — agree that inter-region correlation graphs carry no more next-season information than a random graph of the same density, and the stability analysis supplies the mechanism.

What exists already. Correlation-derived adjacency is routine in epidemic GNN work. Density-matched random-graph controls are not routinely reported alongside it.

Evidence. Tables 7.1, 7.4 and 7.6.

Limit. Ten nodes, six seasons, one disease. This does not show that spatial structure is useless in epidemic forecasting generally — it shows that *this* way of inferring it does not survive a control at this scale.

C5. A bounded negative result on subtype composition for incidence

What it is. At subtype resolution, composition carries no recoverable signal for incidence forecasting at any horizon, with the smallest detectable effect stated (5.5–10.8%).

Why it counts. A null with a working positive control and a stated power floor bounds what a genomic forecasting system can deliver at this resolution. Without the floor it would be uninterpretable.

Limit. It is a statement about subtype composition, not about pathogen genomics in general.

9.2 Potential contributions, not yet demonstrated

- **Clade or antigenic-resolution features.** The incidence null may be a resolution artifact. The pipeline stored metadata only, so this is untested and would require a fresh sequence download.
- **Generalisation beyond US influenza.** Every finding is on one pathogen in one surveillance system.
- **Coverage-conditioned uncertainty.** Explored and *not* supported: stratified conformal intervals were marginally worse than pooled, and coverage's relation to error reverses sign with horizon because coverage is confounded with epidemic phase. Recorded as an attempted direction that did not work.
- **Prospective validation.** All results are retrospective on archived submission dates.

9.3 Claims deliberately not made

- No accuracy superiority over operational systems such as the CDC FluSight ensemble; that comparison was never run and would not be favourable.
- No transmission claims. Genomic similarity between regions is reported as predictive association only.
- No claim that the GNN is competitive. It is not, at this data scale.

- No causal claim about delay. The counterfactual in E3c is a simulation under a stated and unrealistic uniformity assumption.

10. Limitations and threats to validity

- **Sample size.** Six seasons, five folds, 1,980 core region-weeks. This is the binding constraint on every result and is why sign consistency rather than significance testing is the inference tool.
- **Genomic resolution.** Subtype only. Nucleotides were never stored, so clade, antigenic-distance and phylogenetic features are untestable without re-acquisition.
- **Regime change.** One of five folds crosses the COVID discontinuity and behaves differently throughout. It is reported separately rather than dropped.
- **Counterfactual realism.** E3c assumes uniform delay reduction.
- **Single surveillance system.** US HHS regions and CDC FluView only.
- **The NCBI snapshot is not re-fetchable.** GenBank gains records retroactively, so a refetch returns a different dataset. The snapshot *is* the dataset.

11. Reproducibility and artifact index

Every number in this document was computed at build time from the CSVs below; none was transcribed by hand.

experiment	script	result CSVs	figures
E1	`src/analysis/delay_coverage_analysis.py`	coverage_*.csv, delay_summary.csv, coverage_baseline.json	fig_coverage_vs_peak, fig_coverage_crossover, fig_onset_gap, fig_delay_distribution
E2	`src/analysis/baselines.py`	baseline_results.csv, baseline_summary.csv	fig_baseline_by_horizon, fig_baseline_skill_by_fold, fig_baseline_skill_vs_persistence
E3a	`src/analysis/genomic_ablation.py`	ablation_results.csv, ablation_contrasts.csv, ablation_delay_cost.csv	fig_ablation_arms, fig_ablation_by_fold, fig_ablation_headline
E3b	`src/analysis/dominance_forecast.py`	dominance_results.csv, dominance_delay_cost.csv, dominance_decomposition.csv	fig_dominance_headline, fig_dominance_delay_cost, fig_dominance_by_fold
E3c	`src/analysis/delay_policy_surface.py`	policy_surface.csv, policy_surface_signs.csv	fig_policy_surface, fig_policy_mechanism, fig_policy_frontier
E6	`src/analysis/graph_relation_ablation.py`	graph_stability.csv, graph_feature_screen.csv, graph_gnn_ablation.csv	fig_graph_stability, fig_graph_feature_screen, fig_graph_gnn_ablation

Table 11.1 — Artifact index. Regression baselines for E1 live in `coverage_baseline.json`; the script fails loudly on drift.

> **One repository action outstanding.** The title of `fig_ablation_headline.png` predates the E3a null and overstates it; it should be corrected before the figure is reused outside this document, where the caption already notes the discrepancy.